

COURSE MATERIAL	
SUBJECT	BASIC CIVIL AND MECHANICAL ENGINEERING (ME23AES101)
UNIT	1
COURSE	B.TECH
SEMESTER	1-1
DEPARTMENT	MECHANICAL ENGINEERING
PREPARED BY (Faculty Name/s)	Mr. P Karunya Moorthy Assistant Professor
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1. COURSE OBJECTIVES

The objectives of this course is to

1. Get familiarized with the scope and importance of Mechanical Engineering in different sectors and industries.
2. Explain different engineering Materials and different manufacturing processes.
3. Provide an overview of different thermal and mechanical transmission systems and introduce basics of robotics and its applications.

2. PREREQUISITES

Students should have knowledge on

1. Engineering Physics

3. SYLLABUS

UNIT I

Introduction to Mechanical Engineering: Role of Mechanical Engineering in Industries and Society- Technologies in different sectors such as Energy, Manufacturing, Automotive, Aerospace, and Marine sectors.

Engineering Materials- Metals-Ferrous and Non-ferrous, Ceramics, Composites, Smart materials.

4. COURSE OUTCOMES

On completion of the course, the student should be able to

CO1: Understand the different manufacturing processes (L1).

CO2: Explain the basics of thermal engineering and its applications (L2).

CO3: Describe the working of different mechanical power transmission systems and power plants and describe the basics of robotics and its applications (L2).

5. Co-PO / PSO Mapping

CAD/ CAD	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	PO11	PO12	PSO1	PSO2
CO1	3												3	
CO2	3						1						3	
CO3	3												3	

6. LESSON PLAN

LECTURE	WEEK	TOPICS TO BE COVERED	REFERENCES
1	1	Introduction to Basic Mechanical Engineering	T1
2		Role of Mechanical Engineering in Industry and Society	T1
3		Mechanical Engineering Technology in Energy sector	T1
4		Mechanical Engineering Technology in Manufacturing sector	T1, R1
5	2	Mechanical Engineering Technology in Automotive sector	R1
6		Mechanical Engineering Technology in Aerospace and Marine sectors	T1, R1
7		Introduction to Engineering Materials	T1, R1
8		Ferrous and Non Ferrous metals with applications	T1, R1
9	3	Ceramics and Smart Materials	T1, R1
10		Composites	T1, R1

7. ACTIVITY BASED LEARNING

1. Engineer's Day Out – Sector Scavenger Hunt
2. Build-a-Product Challenge

8. LECTURE NOTES

1.1 INTRODUCTION TO MECHANICAL ENGINEERING

Mechanical Engineering is defined as the branch of engineering that deals with the design, development, construction, and operation of mechanical systems and tools. It includes machines, tools, and equipment used in various industries, such as transportation, manufacturing, power generation, and medical devices etc.

Sub disciplines of mechanical engineering: Mechanical engineering is a broad field with many subdiscipline:

Automobile Engineering , Aerospace Engineering , Refrigeration and airconditioning, Mechatronics, Robotics, Machine Design Engineering , Manufacturing Engineering and Industrial Engineering.

1.2 ROLE OF MECHANICAL ENGINEERING IN INDUSTRIES

Mechanical engineering plays a crucial role in various industries due to its broad and versatile nature. Here are some key roles and contributions of mechanical engineering in different industries:

Manufacturing Industry:

Designing and optimizing manufacturing processes to improve efficiency and productivity.

Developing and maintaining machinery and equipment used in manufacturing processes.

Implementing quality control measures to ensure product quality and reliability.

Automotive Industry:

Designing vehicles, engines, and other automotive components for efficiency, safety, and performance.

Conducting research and development to enhance fuel efficiency and reduce emissions.

Improving manufacturing processes to increase production output and reduce costs.

Aerospace Industry:

Designing aircraft, spacecraft, and propulsion systems, considering factors like aerodynamics and structural integrity.

Conducting research and development to enhance fuel efficiency, reduce weight,

and improve safety.

Implementing advanced materials and technologies for aerospace applications.

Energy Sector:

Designing and optimizing energy production systems, such as power plants and renewable energy systems.

Developing technologies for energy storage, transmission, and distribution.

Innovating energy-efficient solutions to reduce environmental impact.

Healthcare Industry:

Designing medical devices and equipment, such as prosthetics, medical imaging machines, and surgical instruments.

Developing and improving manufacturing processes for pharmaceuticals and medical products.

Contributing to the design of facilities and systems to optimize healthcare operations.

Consumer Electronics and Appliances:

Designing and improving consumer electronics, appliances, and gadgets for efficiency, safety, and user-friendliness.

Conducting research and development to enhance the performance and features of electronic devices.

Construction and Building Industry:

Designing heating, ventilation, and air conditioning (HVAC) systems for buildings.

Designing structural components and systems for safe and efficient construction.

Innovating sustainable and energy-efficient building materials and technologies.

Marine and Offshore Industry:

Designing and optimizing marine vessels, offshore structures, and drilling equipment.

1.3 ROLE OF MECHANICAL ENGINEERING IN SOCIETY

Power Generation: Mechanical engineers design and develop power-generating machines such as internal combustion engines, gas turbines, and steam and wind turbines etc.

Heating and Cooling Systems: They design and develop heating, ventilation, refrigeration and air conditioning systems for buildings and other structures.

Transportation: Mechanical engineers are involved in designing and developing transportation systems, including cars, trains, airplanes, steamers and boats.

Industrial Equipment: They design, develop and maintain industrial equipment such as machine tools, robots, and conveyor systems & belts

Agriculture: Mechanical engineers involved in design and development of various agriculture machinery such as harvesters, tractors and irrigation system

Defense: Mechanical Engineers monitor, maintain and repair military hardware such as tanks, guns and logistic vehicles

Mechanical Engineers involved in design, development, inspection and refit of weapon systems

Overall, Mechanical Engineers are involved in designing, building, and maintaining the engines, machines, and structures that make modern life possible and comfortable.

They contribute to society by using their skills to improve the safety, security, efficiency, and comfort of the systems and devices that we rely on every day.

1.4 ROLE OF MECHANICAL ENGINEERING TECHNOLOGY IN ENERGY SECTOR

Energy is the ability to do work

According to the laws of conservation of energy, "energy can neither be created nor destroyed but can only be converted from one form to another".

Mechanical engineers work on the design of alternative energy systems like wind turbines and solar panels, design and optimization of systems for extracting and processing petroleum products, electric power generation and transmission system design, operation, and optimization.

Source of Energy

Fossil Fuels:

Coal: Historically a major source of electricity and industrial heat production, although its use is declining due to environmental concerns.

Oil and Gas: Used for transportation, heating, electricity generation, and various industrial processes

Renewable Energy Sources:

Solar Energy: Joining sunlight through photo voltaic cells or concentrating solar power systems to generate electricity and heat for various applications.

Wind Energy: Capturing wind using wind turbines to generate electricity for both onshore and offshore applications.

Geothermal Energy: Tapping into the Earth's heat to produce electricity and provide heating and cooling in buildings.

Nuclear Energy:

Joining nuclear reactions (fission) to generate a significant amount of electricity, often in large-scale nuclear power plants.

Hybrid and Advanced Energy Technologies:

Hybrid Systems: Combining multiple energy sources to enhance efficiency and reliability, such as combining solar and wind power.

Smart Grids: Implementing advanced digital communication and control systems to enhance the efficiency, reliability, and flexibility of electricity grids..

Ocean Energy: Joining energy from ocean tides, waves, and thermal gradients to generate electricity through technologies like tidal turbines and ocean thermal energy conversion (OTEC).

Waste-to-Energy:

Converting waste materials, such as municipal solid waste or agricultural waste, into energy through incineration, gasification, or anaerobic Digestion

1.5 ROLE OF MECHANICAL ENGINEERING TECHNOLOGY IN MANUFACTURING SECTOR

AI, robots, 3D printing, and the like are all the latest manufacturing trends in manufacturing technology, Robots on the factory floor are growing at warp speed thanks to their declining cost and increasing capabilities.

Technologies in Manufacturing

Manufacturing: something made from raw materials by hand or by machinery.

The 4 types of manufacturing we are going to explore are:

Casting and molding, 2. Machining.3.Joining.4. Shearing and forming.

1. Casting and Molding:

Molding or Mold making is the act of creating the cavity / form that carries a negative or reverse impression of an original model.

Casting is the act of pouring liquid material into the cavity of a mold. After a period of time, this liquid molten metal get solidify .This process is known as a casting.

There are four other types of molding for manufacturing:

Injection molding: which melts plastic to create 3-D materials such as phone cases and laptop covers.

Blow molding: used to make piping and glass objects

Compression molding: used for large-scale products like tires for the car industry

Rotational molding: used for furniture and shipping drums.

2. Machining:

Machining is a process used in the production of many metal products

Machining is a manufacturing process where the desired shape is created by removing material

There are many types of machining process like turning drilling milling grinding planning etc.,

Turning Turning is a machining method that uses a lathe to rotate the metal while a cutting tool moves linearly to remove metal along the diameter, creating a cylindrical shape.

Drilling Drilling is a machining process that uses multi-point drill bits to produce cylindrical holes in the workpiece. In drilling, the machine feeds the rotating drill bit perpendicularly to the plane of the workpiece's surface

Milling Milling is a machining method that uses rotating multi-point cutting tools to remove material from the workpiece.

Grinding Grinding is a machining method that removes small amounts of material

from both flat surfaces and cylindrical shapes.

3. Joining

It is defined as joining of two metal parts either temporarily or permanently with or without application of heat or pressure

Classification

Temporary joining

Fastening

Key joint

Permanent joining

Welding

Soldering

Shearing is the manufacturing process of cutting a piece of material into two separate pieces.

Forming is the manufacturing process whereby, a piece of material is bent into a specific shape.

1.6 ROLE OF MECHANICAL ENGINEERING TECHNOLOGY IN AUTOMOTIVE SECTOR

The automotive sector has seen rapid advancements in technology over the years, transforming the way vehicles are designed, manufactured, operated, and maintained.

Here are some key technologies in the automotive sector:

Electric and Hybrid Vehicles:

Electric vehicles (EVs) and hybrid vehicles use electric power or a combination of electric and internal combustion engine power to reduce emissions and improve fuel efficiency.

Autonomous Vehicles (Self-driving Cars):

Autonomous vehicles use sensors, cameras, radar, lidar, and advanced algorithms to navigate and operate without human intervention. Levels of automation range from driver assistance to fully autonomous systems.

Advanced Driver Assistance Systems (ADAS):

ADAS includes features like adaptive cruise control, lane-keeping assistance, automatic emergency braking, parking assistance, and more to enhance driver safety and convenience.

Cybersecurity Solutions:

Given the connectivity of modern vehicles, cybersecurity measures are essential to protect against cyber threats and ensure the safety and security of both the vehicle and its occupants.

4. Shearing and Forming

Energy-efficient and Lightweight Materials:

The use of lightweight materials like carbon fiber, aluminum, and high-strength steel helps improve fuel efficiency and reduce emissions in traditional combustion engine vehicles.

Advanced Manufacturing Technologies:

Manufacturing processes are evolving with automation, robotics, 3D printing, and AI, enabling faster and more efficient production of automotive components and vehicles.

Vehicle Electrification and Charging Infrastructure:

Expansion of charging infrastructure for electric vehicles, including fast chargers and wireless charging, to support the growing adoption of EVs.

Predictive Maintenance and Telematics:

Telematics systems collect data on vehicle health, driving behavior, and more to facilitate predictive maintenance and optimize vehicle performance.

1.7 ROLE OF MECHANICAL ENGINEERING TECHNOLOGY IN AEROSPACE SECTOR

The aerospace sector has witnessed significant advancements in technology, enabling innovations in aircraft design, space exploration, satellite communications, and air traffic management. Here are some key technologies in the aerospace sector:

Advanced Materials and Composites:

Lightweight and high-strength materials such as carbon fiber composites are widely used in aerospace to improve fuel efficiency and reduce the weight of aircraft.

Jet Propulsion and Engine Technologies:

High-efficiency jet engines and advancements like turbofan and turbojet engines contribute to improved fuel efficiency, reduced emissions, and enhanced performance.

Aircraft Design and Aerodynamics:

Computational Fluid Dynamics (CFD) and other simulation tools aid in optimizing aircraft shapes, wing designs, and aerodynamics for better fuel efficiency and reduced drag.

Satellite Technologies:

Advancements in satellite technology have led to a proliferation of small satellites, improving communication, Earth observation, navigation, and weather forecasting.

Spacecraft and Satellite Propulsion:

Advanced propulsion systems, such as electric propulsion and ion thrusters, provide efficient propulsion for satellites and deep space missions.

Remote Sensing and Earth Observation:

Satellite-based remote sensing technologies monitor and collect data on Earth's surface, oceans, atmosphere, and climate for various applications like environmental monitoring, disaster management, and agriculture.

Air Traffic Management (ATM) and Air Navigation Systems:

Modern ATM systems use automation, artificial intelligence, and advanced communication technologies to enhance airspace efficiency, safety, and reduce congestion.

Environmental Sustainability Technologies:

Research and development of technologies to reduce aviation's environmental impact, including efforts to lower carbon emissions and noise levels.

Cybersecurity for Aerospace:

Implementing robust cybersecurity measures to protect critical aerospace systems, aircraft, satellites, and data from cyber threats.

1.8 ROLE OF MECHANICAL ENGINEERING TECHNOLOGY IN MARINE SECTOR

The marine sector has seen significant technological advancements aimed at improving safety, efficiency, sustainability, and operational capabilities.

Here are some key technologies in the marine sector:

Autonomous ships and Automated Vessels: The development of autonomous ships and automated vessel control systems aims to improve navigation, reduce human error, and enhance overall safety and efficiency in maritime transport.

Fuel-Efficiency and Emission Reduction Technologies: Innovations in engine design, exhaust gas cleaning systems, and alternative fuels (e.g., LNG, hydrogen, ammonia) help reduce greenhouse gas emissions and improve fuel efficiency in ships.

Ship Design and Lightweight Materials: Advanced materials like lightweight composites, aluminum, and high strength steel contribute to the design of more fuel-efficient vessels and reduce their overall weight

Renewable Energy Integration: Implementation of renewable energy technologies such as wind turbines, solar panels, and hydrokinetic devices to power onboard systems and reduce reliance on traditional fossil fuels.

Underwater Robotics and Remote Sensing:

Utilization of underwater drones, remotely operated vehicles (ROVs), and autonomous underwater vehicles (AUVs) for underwater inspections, maintenance, and research.

Safety and Navigation Systems: Integration of advanced navigation systems, collision avoidance technologies, and real-time monitoring to enhance safety and improve navigation in challenging maritime conditions.

Ocean Data Collection and Research Technologies:

Development of autonomous buoys, oceanographic sensors, and research vessels equipped with advanced instruments for collecting data on ocean currents, temperature, salinity, and marine ecosystems.

1.9 INTRODUCTION TO ENGINEERING MATERIALS

Engineering materials refers to the group of materials that are used in the construction of structures and components.

The major classifications of engineering materials include metals, polymers, ceramics, and composites

1.10 CLASSIFICATION OF ENGINEERING MATERIALS

ExtruDesign.com

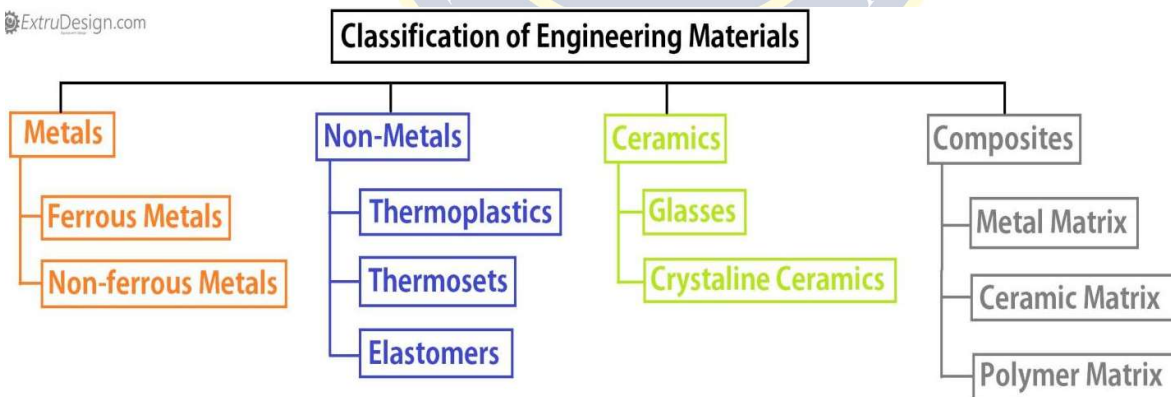


FIG. 1.1 Classification of Engineering Materials

1.11 FERROUS AND NON FERROUS METALS

Ferrous and Non-Ferrous Metals

Ferrous metals are mainly composed of iron and have magnetic properties. Among them, steel is the most widely used metal, in large and small appliances e.g., cars, railways, bridges, household equipment, etc.

Ferrous metals are typically used in the automotive, building, bridge, railway, machinery, shipping, and transportation sectors.

Non - Ferrous Metals

Lead

Uses: •Water pipes, Sheets for roofing and guttering, Accumulator plates in car batteries

Properties: •Sheet metal folds easily and can be soldered

- Castings are brittle and show clearly visible grains
- Resistant to atmospheric conditions

Copper

Uses: •Electrical wiring , Water pipes, Boilers, Rivets

Properties: •Extremely ductile and malleable either hot or cold

- Good conductor of heat and electricity
- Electrical cables can be bent cold
- Melting point 1083°C

Aluminium

Uses: •Foil wrapping for food products, Car body panels ,Aircraft parts – for its lightweight, Cans and Clips

Properties: •Light in weight

- Malleable and ductile
- Resistant to corrosion
- Machines easily
- Difficult to solder and weld

Zinc

Uses: •Protective coating on mild steel (galvanising), Sheets are used for roofing , Zinc often forms protective layer on steel and iron

Properties: •Sheet metal folds easily and can be soldered

- Castings are brittle and show clearly visible grains
- Resistant to atmospheric conditions

Ferrous metals

Cast Iron

Properties •Strong in compression

- Has more carbon than most other types.

Uses Cookware, small components subject to wear such as gears, rods, and pins, and mining equipment.

Wrought Iron

Uses: •Crane hooks, Anchor chains, Cores for electric motors and

transformers

Properties: •Soft, malleable and ductile

•99% iron plus 1% of impurities such including carbon, sulphur, silicon,

Carbon Steel

Properties Typically over 90% iron and the most commonly used ferrous metal.

Uses Structures, furniture, and automotive components.

Stainless Steel

Properties 100% recyclable, Usually have a high amount of chromium that helps resist corrosion better

than carbon steels.

Uses pharmaceutical and medical equipment, food-grade equipment, and knives.

Engineering Steel

Properties Mostly iron, but amounts of copper, vanadium, tungsten, manganese, and other elements can tailor it for toughness, ductility, tensile strength, hardness, and other properties.

Uses Tools, dies, and machining equipment.

Differences between Ferrous Metals and Non Ferrous Metals

S.No.	Ferrous Metals	Non-Ferrous Metals
1	Contain iron as the main element	Do not contain iron as the main component
2	Usually magnetic	Generally non-magnetic
3	Prone to rust and corrosion (without coating)	Resist corrosion and rust naturally
4	Heavier and denser	Lighter in weight
5	High tensile strength	Lower tensile strength than ferrous metals
6	Commonly used in construction and heavy machinery	Used in electrical, aerospace, and decorative applications
7	Recyclable but may degrade in quality	Easily recyclable without much loss of properties
8	Examples: Mild steel, cast iron, stainless steel	Examples: Aluminum, copper, brass, zinc
9	Generally less expensive	Typically more expensive
10	Can spark when struck or ground	Do not spark under normal conditions

1.12 COMPOSITE MATERIALS

Materials having significantly different physical and chemical properties and when combined together produce material with different characteristics from the original constituents.

The 2 essential constituents of composites

1. Matrix composites

2. Reinforcement composites

1. Matrix composites it hold reinforcement by strong bonding

There are 3 types of matrix composites

A. Polymer matrix- high processability

B. Metal matrix - high strength

C. Ceramic matrix - oxidation resistance

2. Reinforcement composites It is the discontinuous phase and gives strength to the composites

It can be classified as following

I. Particle reinforcement composites: as the name suggest that there should be some particles suspended in a matrix. The suspended particles are hard and strong

The load is shared by particle and matrix together

Eg: concrete

Particle reinforcement composites again classified into Large particle reinforced composites and small particle reinforced composites

A) Large particle reinforced composites: The particles which are reinforced inside the matrix are in the form of flakes or particle size is large

Eg: Automobile tyres

B) small particle reinforced composites: particle size is very small like nanosize particles which are added to the matrix which makes composite strong and harder

Eg: Filament of a lamp

II. Fibre reinforcement composites: Here the dispersed phase or the reinforcement is in the form of fibres. These fibres are strong and sticky Fibre reinforcement composites are further classified as continuous fibre reinforced composites and Dis continuous fibre reinforced composites

A) continuous fibre reinforced composites: The fibres are long and continuous in the matrix and are layered parallel to each other

B) Discontinuous fibre reinforced composites: These fibre are short and discontinuous, they may be randomly oriented

III. Structural reinforcement composites: Here the properties of composites depends on structure of composites They are classified as Laminar or layered composites and sandwich composites

A) Laminar or layered composites: It consists of sheets or layers stacked together. As the number of layers increase strength of the composite

increases

Eg: plywood

B) sandwich composites: It consists of a core material sandwiched between two sheets called faces. It is low strength and stiffness.

Face materials may be steel, plywood and Aluminium alloy

Core materials may be synthetic rubber, Wood

Properties of composites:

1. They are Light in weight
2. They have High strength
3. They are corrosion resistant and oxidation resistant
4. They have high toughness

Applications:

1. Automotive parts
2. Marine bodies like boat and ship hulls
3. Storage tanks
4. Plastic pipes
5. Space craft

1.13 CERAMICS

Ceramics (ceramic materials) are non-metallic inorganic compounds formed from metallic or semi-metallic and non-metallic elements.

Atoms of the elements are held together in a ceramic structure by one of the following bonding mechanism: Ionic Bonding, Covalent Bonding, Mixed Bonding (Ionic-Covalent).

Most of ceramic materials have a mixed bonding structure with various ratios between Ionic and Covalent components. This ratio is dependent on the difference in the electro negativities of the elements and determines which of the bonding mechanisms is dominating ionic or covalent.

Glasses: Glasses are a group of ceramics. They are non crystalline

silicates containing oxides such as CaO , Al_2O_3

Salient feature of these materials are their optical transparency and ease of fabrication

Mostly glasses are transformed from a non crystalline state to crystalline by proper heat treatment This process is called crystallization.

It is used in lab equipments like blood sample collector, test tube

Properties:

High mechanical strength

Low coefficient of thermal expansion

Good dielectric properties

Clay products: most widely used ceramic raw material is clay

Clay based products classified into two categories

- i) structural clay products: it include building bricks, tiles, pipes
- ii) Whitewares: it include pottery, porcelain, tableware, plumbing.

1.14. SMART MATERIALS

Smart materials are materials that **respond to external stimuli** (such as heat, light, pressure, electric field, magnetic field, or pH) in a **controlled and reversible manner**.

Characteristics:

Responsiveness: Change properties (shape, color, stiffness, etc.) upon stimulation.

Reversibility: Return to original state after stimulus is removed.

Sensitivity: Detect and react to very small environmental changes.

Applications of smart materials

Smart switches & actuators

Safety device, fuse, alarms

Artificial limbs, blood vessels & muscles

Adhesive tapes/bands

Food packaging industry-wrappers

Smart spoons (Temperature sensitive polymers)

Types of smart materials

Shape Memory Alloys

Piezoelectric Materials

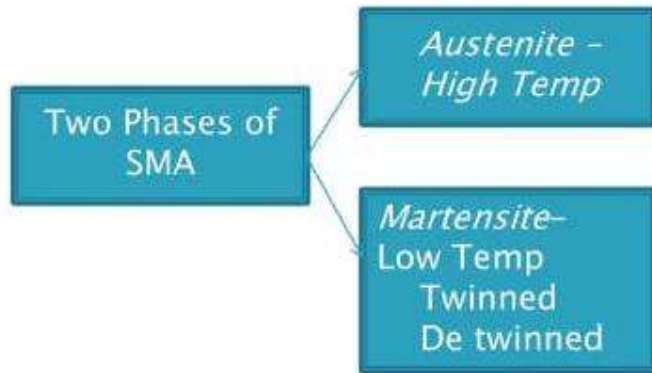
Magnetostrictive Materials

Magneto-Rheological Fluids

Shape memory alloys

SMA's are materials which can revert back to original shape & size on cooling by undergoing phase transformations. Shape Memory Alloys (SMA's) are a unique class of metal alloys that can recover apparent permanent strains when they are heated above a certain temperature.

A phase transformation which occurs between these two phases



Piezoelectric materials

Materials that produce a voltage when stress is applied. (An applied mechanical stress will generate a voltage)

Example: Quartz

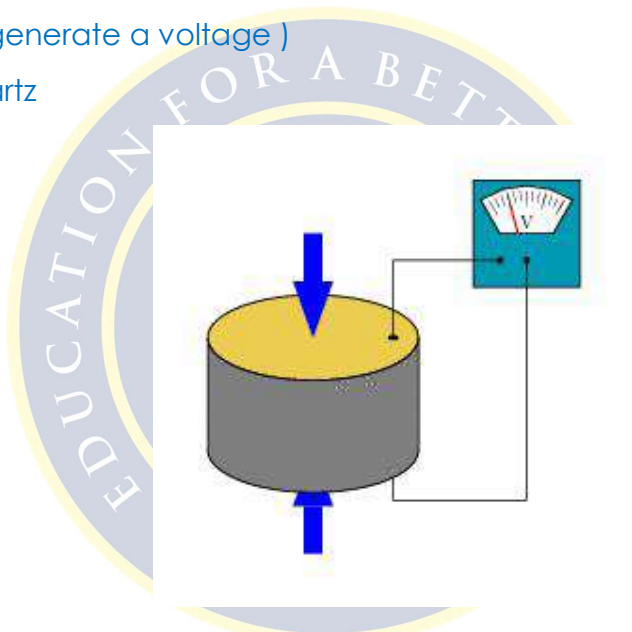


FIG. 1.2 Schematic representation of Piezo electric cell.

The piezoelectric effect describes the relation between a mechanical stress and an electrical voltage in solids.

An LED is wired to a piezoelectric transducer. The LED briefly lights when the device is flicked & shows that electricity has been generated by stress and strain.

Magnetostrictive Materials

Magneto striction is a property of ferro magnetic materials that causes them to change their shape or dimensions during the process of magnetization.

Ex: Fe, Co

Applications

Actuators and Sensors

Magnetostrictive transducers - Convert magnetic energy in to mechanical energy

Vibration Speaker Technology

High quality sound without design compromise. No wires, no boxes, no grilles.

Magneto-Rheological Fluids (MRF)

A MR fluid is a smart fluid which usually consists of 20-40 percent iron particles, suspended in mineral oil, synthetic oil or water.

When subjected to a magnetic field, the magnetic particles inside increase the fluid's viscosity, rendering it to viscoelastic solid.

9. Practice Quiz

1. **What is the primary role of mechanical engineering in the manufacturing industry?**

- A) Financial planning
- B) Product design and process automation
- C) Environmental law enforcement
- D) Marketing strategies

Answer: B

2. **Which of the following sectors heavily utilizes mechanical engineering for propulsion and hydrodynamics?**

- A) Food industry
- B) Marine sector
- C) Fashion industry
- D) Telecommunication

Answer: B

3. **Which of the following is a non-ferrous metal?**

- A) Steel
- B) Cast iron
- C) Copper
- D) Wrought iron

Answer: C

4. What property makes smart materials unique?

- A) High weight and rigidity
- B) Ability to return to original state after a change
- C) Inability to deform
- D) Constant color and temperature

Answer: B

5. Ceramics are best known for which of the following properties?

- A) Excellent electrical conductivity
- B) High thermal conductivity and toughness
- C) Brittleness and high-temperature resistance
- D) Ductility and malleability

Answer: C

6. Shape Memory Alloys are activated by:

- A) Magnetic fields
- B) Light exposure
- C) Temperature change
- D) Voltage supply

Answer: C

7. What type of material is made by combining a matrix and reinforcement phase?

- A) Ceramic
- B) Alloy
- C) Composite
- D) Polymer

Answer: C

8. Which of these industries relies most on lightweight and high-strength materials?

- A) Food processing
- B) Aerospace
- C) Paper production
- D) Construction

Answer: B

9. Which metal is both non-ferrous and lightweight?

- A) Zinc
- B) Iron
- C) Steel
- D) Aluminum

Answer: D

10. Piezoelectric materials convert:

- A) Heat to electricity
- B) Light to heat
- C) Mechanical stress to electrical charge
- D) Sound to mechanical motion

Answer: C

10. Assignments

S.No	Question	BL	CO
1	What are the key roles of Mechanical Engineering in the manufacturing industry?	2	1
2	What are the different types of ferrous metals, explain the basic properties and its applications?	2	1
3	What is a composite? Write the Properties and applications of composites.	2	1
4	Distinguish between the following Ferrous and Non-Ferrous metals.	2	1
5	Explain the contributions of Mechanical Engineering to the welfare of society?	2	1

11. Part A- Question & Answers

S.No	Question& Answers	BL	CO
1	How does mechanical engineering contribute to the manufacturing industry? A: It helps in designing manufacturing systems, automating production lines, and improving the efficiency and quality of mass production.	1	1
2	What is the significance of mechanical engineering in the aerospace industry? A: Mechanical engineers develop lightweight materials, propulsion systems, and control mechanisms for aircraft and spacecraft.	1	1
3	Differentiate between ferrous and non-ferrous metals. A: Ferrous metals contain iron and are magnetic (e.g., steel), while non-ferrous metals do not contain iron and are corrosion-resistant	1	1

	(e.g., aluminum).		
4	Define composite materials and give one example. A: Composites are materials made from two or more different substances to achieve superior properties. Example: Carbon fiber-reinforced polymers.	1	1
5	What are smart materials? Mention one use. A: Smart materials respond to external stimuli like heat, pressure, or electricity. Example: Shape Memory Alloy used in medical stents.	1	1

12. Part B- Questions

S.No	Question	BL	CO
1	Explain the role of Mechanical Engineering in the Aerospace and Marine Sectors? (a)	2	1
2	Explain the concepts that define the role of Mechanical Engineering in Automotive Industry? (a)	2	1
3	What are the different types of ferrous metals, explain the basic properties and its applications?	2	1
4	What is a Metal? Write its classification and mention its properties and applications? (a)	2	1
5	Write a short note on a) Smart materials b) Ceramics	2	1

13. Supportive Online Certification Courses

1. "Basics of Mechanical Engineering" – NPTEL (SWAYAM)

By: IITs/NITs professors

Topics: Thermodynamics, Mechanics, Robotics, Manufacturing, Industrial Engineering and Design.

2. "Introduction to Mechanical Engineering Design and Manufacturing" – Coursera

By: MIT

Topics: CAD, prototyping, mechanical design process

3. "Mechanical Engineering Basics" – Udemy

Covers: Core subjects (fluid mechanics, strength of materials, thermodynamics)

Level: Beginner

14. Real Time Applications

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1	Aerospace Sector Application: Lightweight Structural Components and Jet Engines Use of advanced mechanical systems to reduce weight and improve efficiency. Real Example: Use of composite wings and jet turbine blades in Boeing 787 Dreamliner.	1
	Marine Sector Application: Propulsion Systems and Hull Design Mechanical engineers ensure efficiency, safety, and hydrodynamic performance in marine vessels. Real Example: Submarine pressure hulls and anti-corrosion systems in offshore platforms.	
2	Ferrous Metals Application: Construction beams and car chassis Chosen for their strength and durability. Real Example: Steel used in skyscrapers and rail tracks. Non-Ferrous Metals Application: Electrical wiring, aircraft structures Lightweight and corrosion-resistant. Real Example: Aluminum in aircraft; copper in electrical cables.	1
3	Composites Application: High-strength, low-weight parts in automotive and aerospace Blend of fibers and resins for specific performance. Real Example: Carbon fiber bicycles, Formula 1 car bodies.	1
4	Ceramics Application: Insulators and heat-resistant surfaces Used in electronics and furnaces. Real Example: Spark plugs and ceramic tiles on spacecraft heat shields.	1
5	Smart Materials Application: Actuators, sensors, shape-changing components Respond to stimuli like heat, light, or pressure. Real Example:	1

15. Contents Beyond the Syllabus

Sector	Advanced Topics Beyond the Syllabus
Energy	- Hydrogen fuel cells- Smart grids with AI control- Waste heat recovery in industries
Manufacturing	- Industry 4.0 (IoT in manufacturing)- Additive manufacturing (3D printing)- Digital Twin systems
Automotive	- Autonomous vehicle mechanics- Electric vehicle battery thermal management- Regenerative braking systems
Aerospace	- Hypersonic propulsion systems- Use of shape memory alloys in wing design- AI-based flight control
Marine	- Green shipping technologies- Underwater drones- Wave and tidal energy systems
Material Type	Beyond-the-Syllabus Concepts
Ferrous Metals	- Nano-coated steel for corrosion resistance- Lightweight high-strength steel alloys for space and defense
Non-Ferrous Metals	- Recyclable aluminum alloys for electric vehicle chassis- Graphene-infused copper for higher conductivity
Ceramics	- Transparent ceramics for armor and screens- Ceramic-matrix composites (CMCs) used in jet engines
Composites	- Bio-composites for sustainable construction- 3D woven composites in aerospace- Self-healing composites
Smart Materials	- Magneto-rheological fluids in active suspension systems- Electroactive polymers (EAPs) for soft

16. Prescribed Text Books & Reference Books

Textbooks:

1. An introduction to Mechanical Engg by Jonathan Wicker and Kemper Lewis, Cengage learning India Pvt. Ltd.

Reference Books:

1. 3D printing & Additive Manufacturing Technology- L. Jyothish Kumar, Pulak M Pandey, Springer publications
2. Thermal Engineering by Mahesh M Rathore Tata McGraw Hill publications (India) Pvt. Ltd.
3. G. Shanmugam and M.S.Palanisamy, Basic Civil and the Mechanical Engineering, Tata McGraw Hill publications (India) Pvt. Ltd.

17. Mini Project Suggestion

Mini Project: Automated Conveyor System

Objective:

Design and build a small-scale conveyor belt using a DC motor and simple automation controls to simulate material handling in manufacturing.

Overview:

A conveyor belt system is widely used in manufacturing industries to transport materials from one point to another efficiently. This mini project focuses on building a working model of a conveyor belt powered by a DC motor, with basic auMechanical Design:

Frame Construction:

Build a rectangular frame using wood or metal rods to support the rollers. The length can be around 50-70 cm for easy handling.

Roller Assembly:

Mount two rollers at each end of the frame. One roller is connected to the DC motor via a coupling or belt/pulley system to drive the conveyor belt.

Belt Material:

Use a flexible rubber or fabric strip wrapped around the rollers to act as the conveyor belt. It should be tight enough to avoid slipping but not too tight to strain the motor.

Motor Mounting:

Fix the DC motor on the frame securely so its shaft aligns with the driving roller.

Connect the motor shaft to the roller using a coupling or friction fit.

Automation Features (Optional):

Start/Stop Control:

Use push buttons connected to Arduino to control the motor.

Speed Control:

Use PWM (Pulse Width Modulation) to vary motor speed via the microcontroller.

Object Detection:

Integrate an IR sensor or ultrasonic sensor to detect objects on the belt and automatically stop the motor.

Manufacturing Process Simulation:

Use small objects (like beads or cubes) to simulate material movement on the conveyor.

Observe and analyze speed, load capacity, and smoothness of the belt.

Adjust tension and motor speed for optimal performance.

Learning Outcomes:

Mechanical Design:

Understanding frame design, roller mechanics, and belt tension.

Motor Control:

Basics of DC motor operation, power supply considerations, and simple ON/OFF control.

Automation:

Introduction to microcontroller-based motor control and sensor integration.

Manufacturing Insight:

Real-world application of conveyors in factories, highlighting efficiency and automation benefits.

Steps to Build:

Construct the frame and install rollers.

Attach the belt and adjust tension.

Mount the motor and connect to the driving roller.

Wire the motor to the power supply and switch.

Test manual ON/OFF operation.

(Optional) Program the Arduino for automated control.

Test with objects on the belt and observe performance.

